

求积分

$$\begin{aligned}& \int_0^{\pi} \frac{x \sin x}{1 - 2r \cos x + r^2} dx \\&= \frac{1}{2} \int_{-\pi}^{\pi} \frac{x \sin x}{1 - 2r \cos x + r^2} dx \\&= \frac{1}{4i} \int_{-\pi}^{\pi} \frac{x(e^{ix} - e^{-ix})}{(r - e^{ix})(r - e^{-ix})} dx \\&= \frac{1}{2i} \int_{-\pi}^{\pi} \frac{x e^{ix}}{(r - e^{ix})(r - e^{-ix})} dx\end{aligned}$$

令 $z = e^{ix}$ 则 $dx = \frac{1}{iz} dz$

得到

$$\begin{aligned}&= -\frac{1}{2i} \oint \frac{\ln z}{(r - z)(r - \frac{1}{z})} dz \\&= -\pi \operatorname{Res}\left[\frac{\ln z}{(r - z)(r - \frac{1}{z})}\right] = \frac{\pi r \ln r}{|r^2 - 1|}\end{aligned}$$

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